



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 11

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Program:

```
#include <stdio.h>
```

```
void heapify(int a[], int n, int i)
```

```
{
```

```
    int largest = i;
```

```
    int left = 2 * i + 1;
```

```
    int right = 2 * i + 2;
```

```
    int temp;
```

```
    if (left < n && a[left] > a[largest])
```

```
        largest = left;
```

```
    if (right < n && a[right] > a[largest])
```

```
        largest = right;
```

```
    if (largest != i)
```

```
    {
```

```
        temp = a[i];
```

```
        a[i] = a[largest];
```

```
        a[largest] = temp;
```

```
    heapify(a, n, largest);
```

```
    }
```

```
}
```

```
void heapSort(int a[], int n)
```

```
{
```

```
    int i, temp;
```

```
    for (i = n / 2 - 1; i >= 0; i--)
```

```
        heapify(a, n, i);
```

```
    for (i = n - 1; i > 0; i--)
```

```
    {
```

```
        temp = a[0];
```

```
        a[0] = a[i];
```

```
        a[i] = temp;
```

```
        heapify(a, i, 0);
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
    int a[] = {4, 10, 3, 5, 1};
```

```
    int n = 5, i;
```

```
    printf("Name: Dharshana Sivakumar\n");
```

```
    printf("Heap Sort\n\n");
```

```
    heapSort(a, n);
```

```
    printf("Sorted Array: ");
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        printf("%d", a[i]);
```

```
        if (i < n - 1)
```

```
            printf(", ");
```

```
    }  
    printf("]");  
    return 0;  
}
```

Output:

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 3 void heapify(int a[], int n, int i) 4 { 5 int largest = i; 6 int left = 2 * i + 1; 7 int right = 2 * i + 2; 8 int temp; 9 10 if (left < n && a[left] > a[largest]) 11 largest = left; 12 13 if (right < n && a[right] > a[largest]) 14 largest = right; 15 16 if (largest != i) 17 { 18 temp = a[i]; 19 a[i] = a[largest]; 20 a[largest] = temp; 21 22 heapify(a, n, largest); 23 } 24 } 25 26 void heapSort(int a[], int n) 27 { 28 int i, temp; 29 30 for (i = n / 2 - 1; i >= 0; i--) 31 heapify(a, n, i); 32 33 for (i = n - 1; i > 0; i--)</pre>		<pre>Name: Dharshana Sivakumar Heap Sort Sorted Array: [1, 3, 4, 5, 10]</pre>